

C1MAT1T1 – O – 25 – 4064



FIRST SEMESTER B.SC. (REVISED NEP) DEGREE
EXAMINATION, JANUARY 2026
MATHEMATICS
Algebra – I and Calculus – I (DSC)

Time : 3 Hours]

[Max. Marks : 80

Instruction : Answer **all** questions.

I. Answer **all** questions :

(10×2=20)

- 1) State any two elementary row transformations.
- 2) State Cayley Hamilton theorem.
- 3) Discuss the continuity of the function $f(x) = \begin{cases} 1+x, & x \leq 2 \\ 5-x, & x > 2 \end{cases}$ at $x = 2$.
- 4) Evaluate $\lim_{x \rightarrow 0} \left(\frac{x - \tan x}{x^3} \right)$.
- 5) Find ϕ for the curve $r^2 = a^2 \cos 2\theta$ at $\theta = \frac{\pi}{6}$.
- 6) Define concavity and convexity of a curve $y = f(x)$.
- 7) When the curve $f(x, y) = 0$ is symmetric about the line $y = x$?
- 8) Find the n^{th} derivative of $y = e^{ax+b}$.
- 9) State Lagrange's mean value theorem.
- 10) Expand $\cos x$ in powers of x by Maclaurin's theorem.

II. Answer **any six** questions :

(6×5=30)

- 11) Find the rank of the matrix $\begin{pmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 5 \\ 1 & 2 & 3 & 4 \\ 1 & -2 & 6 & -7 \end{pmatrix}$.

- 12) For what values of λ and μ the system of equations
 $x + 2y + 3z = 5$
 $x + 3y - z = 4$
 $x + 2y + \lambda z = \mu$ have
a) No solution
b) Infinite solution
c) Unique solution ?

[P.T.O.]



- 13) If $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$, then prove that $\lim_{x \rightarrow a} [f(x) + g(x)] = l + m$.
- 14) Show that the curves $r = a(1 + \sin\theta)$ and $r = b(1 - \sin\theta)$ cut orthogonally.
- 15) A function $f(x)$ is continuous in a closed interval $[a, b]$, then prove that $f(x)$ attains its bounds at least once.
- 16) Derive the formula for the radius of curvature of the curve in cartesian form.
- 17) State and prove Cauchy's mean value theorem.
- 18) Find the n^{th} derivative of $e^{ax}\cos(bx + c)$.

III. Answer **any three** full questions :

(3×10=30)

- 19) a) Reduce the matrix $\begin{pmatrix} 1 & 0 & 1 & 1 \\ -1 & -2 & 1 & 3 \\ 3 & 2 & 1 & -1 \\ 1 & 4 & 5 & 9 \end{pmatrix}$ to the normal form and then find its rank.

b) Verify Cayley Hamilton theorem for the matrix $\begin{pmatrix} 2 & 4 \\ 3 & 1 \end{pmatrix}$.

- 20) a) State and prove intermediate value theorem.

b) Evaluate $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x}{2} \right)^{\frac{1}{x}}$.

- 21) a) With usual notation prove that $\tan\phi = r \frac{d\theta}{dr}$.

b) Find the pedal equation of the curve $r = a(1 - \cos\theta)$.

- 22) a) State and prove Leibnitz's theorem to find the n^{th} derivative of product of two functions.

b) If $y = e^{m \sin^{-1} x}$ then prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$.



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[Max. Marks : 80

Instruction : Answer all questions.

I. Answer all questions.

(10×2=20)

1) Define rank of matrix.

2) State Cayley-Hamilton theorem.

3) Discuss the continuity of the function.

$$f(x) = \begin{cases} 3x-2 & \text{if } x \leq 0 \\ x+1 & \text{if } x > 0 \end{cases}$$

at a point $x = 0$.

4) Evaluate $\lim_{x \rightarrow 0} \left(\frac{x - \sin x}{x^3} \right)$.

5) Define curvature of a curve at any point on it.

6) Find the asymptotes parallel to x and y -axes of the curve.

$$x^2y^2 - y^2 = a(x^2 + y^2)$$

7) When the curve $f(x, y) = 0$ is symmetric about x - axis ?

8) Find the n^{th} derivative of $\cos(ax + b)$.

9) State Rolle's theorem.

10) Expand e^x in powers of x by Maclaurin's theorem.

II. Answer any six questions.

(6×5=30)

11) Find the rank of the matrix,

$$\begin{bmatrix} 3 & -1 & -1 & 2 \\ 2 & 2 & 1 & -1 \\ 1 & -3 & 2 & -1 \\ 1 & -3 & 6 & -1 \end{bmatrix}$$

and hence reduce it into normal form.



12) For what values of λ and μ the system of equations.

$$x + y + z = 6$$

$$2x + 4y + 6z = 20$$

$$x + 2y + \lambda z = \mu$$

- a) a unique solution
- b) Infinite solution
- c) No solution.

13) If $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$ then prove that $\lim_{x \rightarrow a} [f(x) + g(x)] = l + m$.

14) Prove that every continuous function $f(x)$ in closed interval $[a, b]$ is bounded in that interval.

15) With usual notations prove that $\tan \phi = r \frac{d\theta}{dr}$.

16) Prove that the curves $r = a(1 - \cos \theta)$ and $r = b(1 + \cos \theta)$ cuts orthogonally.

17) State and prove Lagranges mean value theorem.

18) Find the n^{th} derivative of $e^{ax} \sin(bx + c)$.

III. Answer any three full questions.

(3×10=30)

19) a) Find the rank of the matrix,

$$\begin{bmatrix} -2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

and hence reduce into Echelon form.

b) Find the inverse of the square matrix $\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ using Cayley-Hamilton theorem.

20) a) State and prove intermediate value theorem.

b) Evaluate $\lim_{x \rightarrow 0} (\cos x)^{\frac{1}{x}}$.

21) a) Derive the formula for the radius of curvature of the curve in Cartesian form.

b) Find the pedal equation of the curve $2a = r(1 + \cos \theta)$.

22) a) State and prove Leibnitz theorem to find the n^{th} derivative of product of two functions.

b) If $y = (\sin^{-1} x)^2$ then prove that $(1 - x^2) y_{n+2} - (2n + 1)x y_{n+1} - n^2 y_n = 0$.